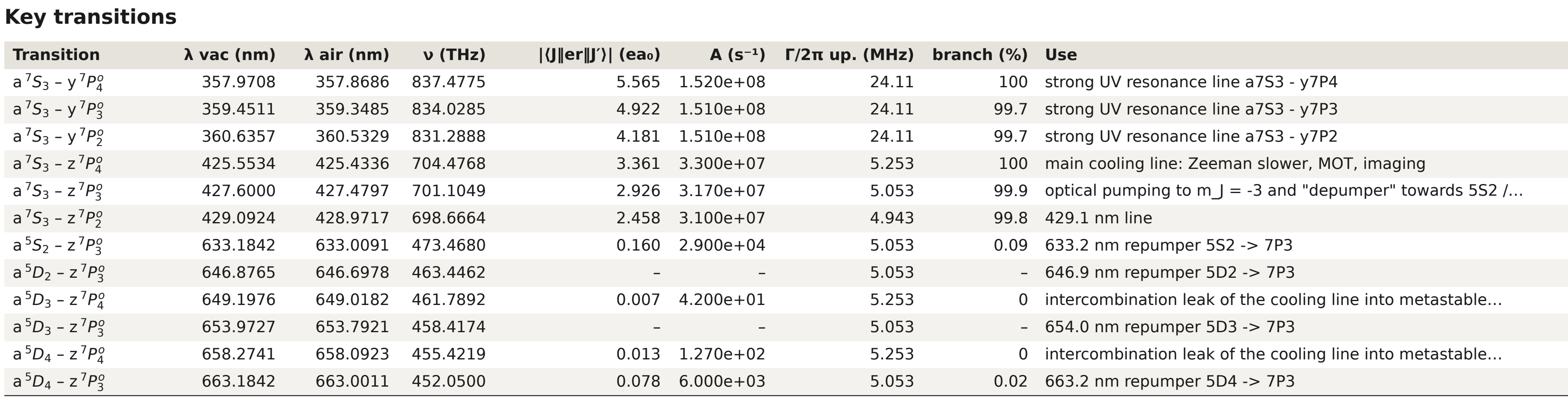


Cr⁺ ionisation limit 54575.60 cm⁻¹ = 6.76651 eV (183.232 nm)

Arrow colour = vacuum wavelength (nm)



300 400 500 600 700 800 1000 1200 1500 2000



3C hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)					
$a_1^+S_3$	-83.5990(20)		F=3/2: +501.59	F=5/2: +292.60	F=7/2: -0.00	F=9/2: -376.20		
$a_1^+D_3$	-173.820		F=3/2: +722.19	F=5/2: +521.46	F=7/2: +86.91	F=9/2: -521.46		
$a_2^+S_3$	-35.68(5)	15.56(5)	F=3/2: +223.44	F=5/2: +122.56	F=7/2: -7.78	F=9/2: -156.68		
$a_2^+D_3$	-48.75(5)	63.02(10)	F=5/2: +396.61	F=7/2: +182.64	F=9/2: -51.95	F=11/2: -276.77		
$z_1^+P_3^0$	-1.5(20)	-2.000	F=3/2: +7.40	F=5/2: +5.55	F=7/2: +1.00	F=9/2: -7.25		
$z_1^+P_3^2$	-11.8(4)	-0.6(6)	F=5/2: +88.20	F=7/2: +47.32	F=9/2: -5.64	F=11/2: -70.95		

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$a^7S_3 - z^7P^o_2$	425.553	53-52	10.7(17)
$a^7S_3 - z^7P^o_0$	427.600	53-52	18(23)
$a^5S_2 - z^7P^o_3$	633.184	53-52	-155(53)
$a^3D_3 - z^7P^o_3$	653.973	53-52	670(28)
$a^3D_4 - z^7P^o_3$	663.184	53-52	819(28)

Shift = v (first isotope) - v (second isotope).